Structure-Aware Neural Decoding for the X-Cube Fracton Code

Finite-Size Results for Logical-Parity Inference

Brian Guarraci
Galatheus
brian@galatheus.com

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Abstract

We study learned logical recovery for the X-cube fracton code under code-capacity X noise with perfect syndrome measurements. A weight-shared 3D convolutional network reads the cube (Z) syndrome and uses a line-aggregation head (AxisLineonHead) to predict the $6L-3$ logical-Z parities in a fixed axis-aligned basis. Models are trained separately at each code size and physical error rate. Across $L \in \{3,4,6,8,10\}$ and $p \in \{0.03, 0.05\}$, three-seed evaluations on disjoint test splits give lower block failure and mean per-sector logical error than the all-zero baseline, BP+OSD-0, and our simplified Brown–Williamson-inspired implementation (BW-naive). At $p = 0.03$, mean per-sector error decreases monotonically from 0.042 to 0.003 over the tested sizes. Block failure increases from 0.306 at $L = 3$ to 0.343 at $L = 4$, then decreases to 0.118 at $L = 10$. At $p = 0.05$, block failure instead increases with size, despite the improvement in per-sector error from $L = 4$ onward. A single-seed $L = 3$ ablation finds comparable performance for line-aggregation and coordinate-convolution heads; global pooling performs like the all-zero baseline. This supports retaining spatial information but does not isolate line aggregation as the unique mechanism. The results demonstrate finite-size learned recovery in the tested regime. They do not establish a threshold, superiority to a validated implementation of the published Brown–Williamson decoder, cross-size transfer, or a passive quantum memory.

1 Introduction

This work began with a classical communication question: if a receiver already has a useful model of a source, how much instance-specific evidence must a sender transmit? A syndrome can identify a set of compatible signals, while a shared prior ranks or samples plausible reconstructions within that set. The companion occupancy-grid work investigates this idea using learned spatial priors, with spectral and coarse side information in the reported experiments. Its objective is reconstruction quality under a bit budget.

Quantum error correction poses a related inference problem with a different target. A syndrome constrains the physical errors that may have occurred; the noise model supplies their prior probabilities. Successful recovery requires the correct logical class, rather than a plausible replacement for an unknown quantum state. This distinction is essential: classical perceptual plausibility is not a quantum recovery criterion.

Here we study that inference task for the X-cube code [1]. Its cube syndrome has a regular three-dimensional representation, while the logical labels in our chosen basis are associated with wrapping lines. We test a convolutional decoder that preserves this spatial organization through a line-aggregation readout. Existing code-specific decoders exploit the same code's planar constraints through matching [5]; a single surface-code-style matching problem over pairs of defects does not directly represent the four-defect patterns of the channel studied here. The overlapping weight-12 cube checks also make this a useful setting in which to compare with a generic BP+OSD-0 baseline.

Contribution and scope. We report three-seed finite-size measurements for code-capacity X noise, using both whole-block failure and mean error over a fixed logical basis. The learned decoder improves on the implemented baselines at every tested operating point. Its block failure decreases from $L = 4$ to $L = 10$ at $p = 0.03$ but increases over that range at $p = 0.05$. Models are trained separately at each (L, p); the results do not measure transfer between sizes or an asymptotic threshold. The head ablation is evidence about retaining position information, with stronger mechanism tests left open.

The companion gauge note characterizes the classical axial-parity codespace, and the surface-code paper studies neural decoding budgets. These motivate a shared question about useful inductive biases; neither the classical theorem nor a physical equivalence between these systems is required for the X-cube benchmark. Passive storage and self-correction remain separate questions about Hamiltonian dynamics and memory lifetime.

2 Background: the X-cube code and the decoding task

The X-cube code [1] places its qubits on the $3L^3$ edges of a periodic $L \times L \times L$ cubic lattice. The stabilizer group is generated by (i) X-type "cross" operators — per vertex, the four-edge planar cross in each of the xy, yz, xz planes (two independent per vertex, since the third is their product), and (ii) Z-type "cube" operators — the product of the twelve edges of each elementary cube. The code is CSS ($H_X H_Z^\top = 0$ over $\mathbb{F}_2$) and we verify $k = N - \mathrm{rank}(H_X) - \mathrm{rank}(H_Z) = 6L-3$ for all simulated sizes $L \in \{2, 3, 4, 5, 6, 8, 10\}$ (giving $k = 9, 15, 21, 27, 33, 45, 57$), matching the known ground-state degeneracy on the 3-torus.

We study the code-capacity bit-flip (X) channel at rate p, detected by the cube (Z) stabilizers. The cube syndrome has exactly L^3 bits — one per cube center — so it is naturally an $L \times L \times L$ scalar field, a direct fit for a 3D convolutional decoder with no temporal axis. The decoding label is the vector of $6L - 3$ logical-Z parities of the error, $\ell_j = \langle \mathrm{logical}_Z^{(j)}, e \rangle \bmod 2$; the decoder predicts these per-sector classes from the syndrome field. We report *block failure* (any sector wrong) and the mean *per-sector logical error rate* (LER).

Why this is decoding, not classification. Predicting the $6L-3$ logical-Z parities is the full decoding output, not a proxy. Given a syndrome s, fix once a syndrome-consistent representative ("pure error") $R(s)$ derived from H_Z; the residual $e \oplus R(s)$ then lies in the logical+stabilizer group, so the only remaining freedom is which logical coset it occupies. In the experiments below the network is trained to predict the raw logical parities $\langle \mathrm{logical}_Z^{(j)}, e \rangle$ of the sampled error. Equivalently, an explicit recovery using an arbitrary $R(s)$ applies the logical correction with parity $\hat{\ell}_j \oplus \langle \mathrm{logical}_Z^{(j)}, R(s) \rangle$; if $R(s)$ is chosen with zero logical-Z parities this offset vanishes. A per-sector logical error occurs precisely when the predicted parity is wrong. This defines a logical-class decoder. The implemented loss is a sum of binary cross-entropies, and each output is thresholded separately. At the population optimum, these outputs estimate the marginal probabilities $P(\ell_j = 1 \mid s)$; their separate modes minimize expected per-sector error. They do not in general maximize the joint probability $P(\ell \mid s)$, which is the decision rule for minimizing block failure. We therefore report block failure separately and do not identify this training objective with exact joint maximum-likelihood coset decoding.

3 Classical motivation and the algebraic connection

The classical axial-parity result is $\ker H = \mathrm{image}(\partial_x \partial_y \partial_z)$ over $\mathbb{F}_2$ on a periodic n^3 voxel lattice, with $\mathrm{rank}(H) = 3n^2 - 3n + 1$. Writing $T = \partial_x \partial_y \partial_z$, a voxel residual can be parameterized as $T\phi$, with the redundancy $\phi \mapsto \phi + \kappa$ for $\kappa \in \ker T$. This is an exact statement about the classical constraint set and its parameterization.

Subsystem constraints and redundant potentials also occur in fracton models [1, 3]. The classical identity alone does not identify the codec with a particular quantum gauge theory or the X-cube phase: that would require a mapping of local degrees of freedom, operators, and physical constraints. No such identification is assumed here. What carries across the paper family is a statistical question: how should a decoder combine a prior with syndrome evidence while preserving the geometry relevant to its target?

4 Decoder and baselines

AxisLineonHead. The decoder is a weight-shared 3D residual CNN (default width 64, depth 8) over the L^3 cube-syndrome field, followed by a line-aggregation head: features are collapsed along each logical-line axis, passed through a 2D transverse convolution, and gathered into the $6L - 3$ per-sector logits. The head is the only code-aware component; the body is a generic 3D CNN. Training uses $3{\times}10^5$ samples per cell, 30 epochs, a cosine learning-rate schedule, and batch size 2048 for $L \leq 6$, reduced to 768 at $L = 8$ and 384 at $L = 10$ for memory.

Baselines. (i) *Trivial*: predict the all-zero logical class (the per-sector majority prediction for the channel and logical basis used here), a reference for prediction without syndrome information. (ii) *BP+OSD-0*: belief propagation with order-0 ordered-statistics post-processing on H_Z, a standard general-purpose qLDPC decoder [12]; we report its block failure and BP convergence fraction. (iii) *BW-naive*: our simplified, naive-weight implementation inspired by Brown–Williamson's X-cube matching framework, which exploits the rigid fracton structure to reduce decoding to two-dimensional matching problems [5]. We implement the cube-defect ("planeon") matching variant (per-plane MWPM + cluster neutralization; 100% syndrome-consistent, exact on isolated single-edge errors); BW's BP edge-reweighting that lifts its threshold 3.7% $\rightarrow$ 4.3% is a documented extension we do not implement, so we do not claim to benchmark the fully tuned Brown–Williamson decoder. Our implementation passes syndrome-consistency and isolated-single-error checks, but these do not establish fidelity to the published algorithm's logical performance. Its poor large-size behavior warrants an independent audit of periodic wrapping, cluster neutralization, and logical-class selection. Comparisons here concern the released BW-naive implementation; we do not attribute its failure rates to the published decoder.

Finite-data local lookup control (Gate-0). As a locality control, we evaluate a held-out local-pattern lookup over the syndrome in a *radius-1* transverse neighborhood of each logical line. This empirical frequency estimator tests one finite-data use of that window; it is a control for the mechanism analysis, not a formal Bayes-optimal bound and not a bound over local decoders of arbitrary receptive field. We run this control through $L = 6$; since the radius-1 advantage is already $\leq 0.3\%$ there, we retain it as a limited control rather than re-running it for the $L = 8, 10$ rows. Failure of this lookup can also reflect sparse training coverage of patterns; it does not prove an absence of local information.

Protocol. Every cell is run over seeds $\{42, 43, 44\}$ with a disjoint test sampler seed offset from train and validation. We report seed-bootstrap 95% confidence intervals on block failure and paired McNemar tests (on identical shots) for learned-vs-trivial and learned-vs-BP+OSD-0. The predeclared win condition (fixed before evaluation) is: learned block-failure CI upper bound below the trivial CI lower bound at $L = 3$, $p = 0.05$. All numbers come from one reproducible pipeline built on `reproduce_xcube_results.py`: a single command for $L \leq 6$, and, for $L \geq 8$, per-cell shards of that same script merged with a parallel CPU BW-naive run. Exact commands and output paths are in `REPRODUCIBILITY.md`.

Table 1: Block failure and per-sector LER (3-seed means) for the learned decoder vs. our naive-weight Brown–Williamson decoder ("BW-naive"), BP+OSD-0, and the trivial floor, on the code-capacity X-channel. Lower is better. **The learned decoder wins at every cell on both metrics.** Learned/BW-naive block-failure 95% CIs and exact paired-McNemar counts are in App. A; McNemar $p < 10^{-12}$ ($b \gg c$) for learned-vs-trivial and learned-vs-BP+OSD-0 at every cell, and for learned-vs-BW-naive at $L \leq 6$ (at $L \geq 8$ BW-naive fails near-totally, block failure ≥ 0.99, and is compared by aggregate failure rate, not paired shot-by-shot). The $L = 8$ rows use a memory-reduced batch (768 vs 2048) and the $L = 10$ rows batch 384. The learned, trivial, and BW-naive decoders are evaluated on 40,000 test shots per seed for $L \leq 6$ and 30,000 per seed for $L \geq 8$; the much slower BP+OSD-0 baseline is evaluated on a subset (2,000 shots per seed through $L = 8$, 300 at $L = 10$), with BW-naive run on dedicated CPU nodes and merged. BW-naive (§4) has lower block failure than the trivial baseline before saturation, but higher per-sector error in several cells; these metrics need not rank decoders identically.

		Block failure				Per-sector LER			
L	p	Learned	BW-naive	BP+OSD-0	Triv.	Learned	BW-naive	BP+OSD-0	Triv.
3	0.03	**0.306**	0.391	0.372	0.735	**0.042**	0.098	0.110	0.085
3	0.05	**0.626**	0.703	0.688	0.887	**0.104**	0.204	0.228	0.136
4	0.03	**0.343**	0.682	0.465	0.914	**0.035**	0.169	0.140	0.110
4	0.05	**0.724**	0.937	0.840	0.981	**0.110**	0.301	0.316	0.172
6	0.03	**0.282**	0.935	0.595	0.996	**0.018**	0.242	0.233	0.155
6	0.05	**0.800**	0.999	0.979	1.000	**0.104**	0.392	0.451	0.234
8	0.03	**0.194**	0.993	0.864	1.000	**0.008**	0.295	0.361	0.195
8	0.05	**0.850**	1.000	1.000	1.000	**0.095**	0.441	0.492	0.285
10	0.03	**0.118**	1.000	0.978	1.000	**0.003**	0.336	0.427	0.231
10	0.05	**0.863**	1.000	1.000	1.000	**0.080**	0.466	0.494	0.326

5 Results

The decoder wins at every operating point. Table 1 gives the full grid. At every (L, p) the learned decoder has strictly lower block failure than BW-naive, BP+OSD-0, and trivial, with non-overlapping seed CIs against trivial in the JSON and paired McNemar $p < 10^{-12}$ against trivial and BP+OSD-0 at every cell, and against BW-naive for $L \leq 6$ (where the paired test is tabulated; BW-naive is compared by aggregate failure rate at $L \geq 8$, see App. A) (e.g. at $L = 3, p = 0.03$, learned-vs-trivial b=53294, c=1794; learned-vs-BP+OSD-0 b=511, c=99; learned-vs-BW-naive b=12648, c=2499). BP+OSD-0 itself beats trivial at small codes but degrades with size and noise; its BP convergence fraction falls from ~ 0.8 ($L = 3, p = 0.03$) to ~ 0.03 ($L = 6$, $p = 0.05$). The measured BW-naive gap widens with size; for example, at $L = 6, p = 0.03$ its per-sector error is 0.242, compared with 0.018 for the learned decoder. This is an implementation-level comparison, not a measurement of the published Brown–Williamson algorithm's performance.

Finite-size behavior depends on the metric and noise rate. At $p = 0.03$, improvement is observed on both metrics (Fig. 1). On the *basis-invariant* block-failure rate (all $6L - 3$ sectors correct at once), the learned decoder improves with size at $p = 0.03$ from 0.343 at $L = 4$ to 0.118 at $L = 10$ (the $L = 3$ value 0.306 is the only non-monotone step), while every baseline saturates toward 1. Per logical qubit — the per-sector LER, reported in the frozen axis-aligned wrapping-line logical-Z basis the model is trained against, and thus *basis-relative* (block failure is the basis-invariant metric) — the learned decoder improves *monotonically* from $0.042 \to 0.035 \to 0.018 \to 0.008 \to 0.003$ as $L = 3 \to 4 \to 6 \to 8 \to 10$, while the trivial baseline *worsens* ($0.085 \to 0.110 \to 0.155 \to 0.195 \to 0.231$) and BP+OSD-0 worsens ($0.110 \to 0.140 \to 0.233 \to 0.361 \to 0.427$). At $L = 6$ this is a 9× per-sector advantage over

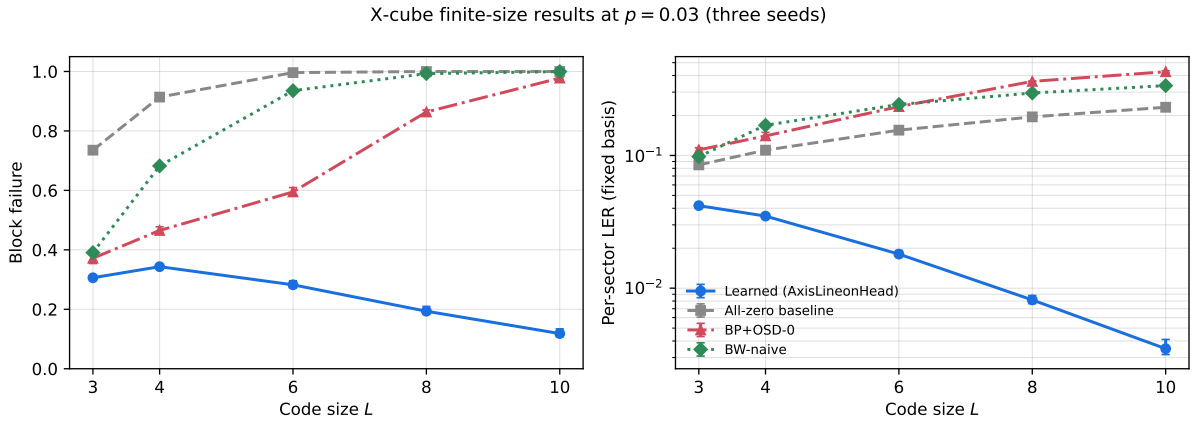


Figure 1: Finite-size performance at $p = 0.03$ (3-seed means and seed-bootstrap 95% intervals; Table 1). Left: block failure, which improves for the learned decoder from $L = 4$ onward. Right: mean per-sector error in the fixed logical basis, which improves at every tested size. Each size has a separately trained model. Both panels use the retained result JSONs without new experiments.

trivial, widening to $\sim$24$\times$ at $L = 8$ and $\sim$66$\times$ at $L = 10$. Strikingly, by $L = 8$ both BP+OSD-0 and BW-naive have per-sector LER *above* the trivial floor, so the learned decoder is the only one of the four that improves on doing nothing *on the per-sector metric*; its learned-vs-trivial McNemar at $L = 8, p = 0.03$ has $c = 0$ — across 72,574 discordant shots the trivial decoder never once beat it. This persists at $L = 10$, where the learned decoder is again the only one of the four below the trivial floor (per-sector 0.003 vs. 0.336 BW-naive, 0.427 BP+OSD-0, 0.231 trivial) and again has $c = 0$ across 79,351 discordant shots. At $p = 0.05$, however, block failure increases from 0.626 to 0.863 over $L = 3$ through $L = 10$, even though per-sector error improves from $L = 4$ onward. Because the number of logical labels grows with L and their errors can be correlated, per-sector improvement alone is insufficient to establish improved whole-block recovery.

Noise dependence and the unresolved crossing. Figure 2 plots the measured dependence on p. For $L \geq 4$, larger sizes have lower block failure at $p = 0.03$ and higher block failure at $p = 0.05$. This reversal motivates a denser noise sweep. Except at $L = 4$, however, only those two physical error rates were measured. Lines between them are visual guides and do not establish a threshold location or an asymptotic finite-size scaling law. A comparison with the published Brown–Williamson or optimal-decoder thresholds requires a resolved sweep, matched error sectors, and compatible failure criteria [5, 4]. The $L = 4$ high-noise sweep measures a single size's advantage over a baseline; its disappearance cannot locate a code threshold.

Ablations and limits of the mechanism evidence. At $L = 3, p = 0.03$, seed 42, the global-pooling head has the same measured block failure as the all-zero baseline (0.7342), whereas the coordinate convolution and line-aggregation heads achieve 0.3022 and 0.3026 (Table 2). These measurements support preserving spatial information in the readout. They do not show a unique advantage from line aggregation, nor separate all differences in capacity and optimization. Matched-capacity, multi-seed ablations at larger sizes are needed to test whether explicit line aggregation offers an additional benefit.

The held-out radius-1 lookup has per-sector advantage over trivial of 34.2% at $L = 3, p = 0.03$, falling to 5.7% at $L = 4, p = 0.03$, 0.1% at $L = 4, p = 0.05$, and $\leq 0.3\%$ for all $L = 6$ cells. The network outperforms this finite-data control. Since the lookup is neither a Bayes-optimal estimator nor an exhaustive family of local decoders, this does not prove that all useful

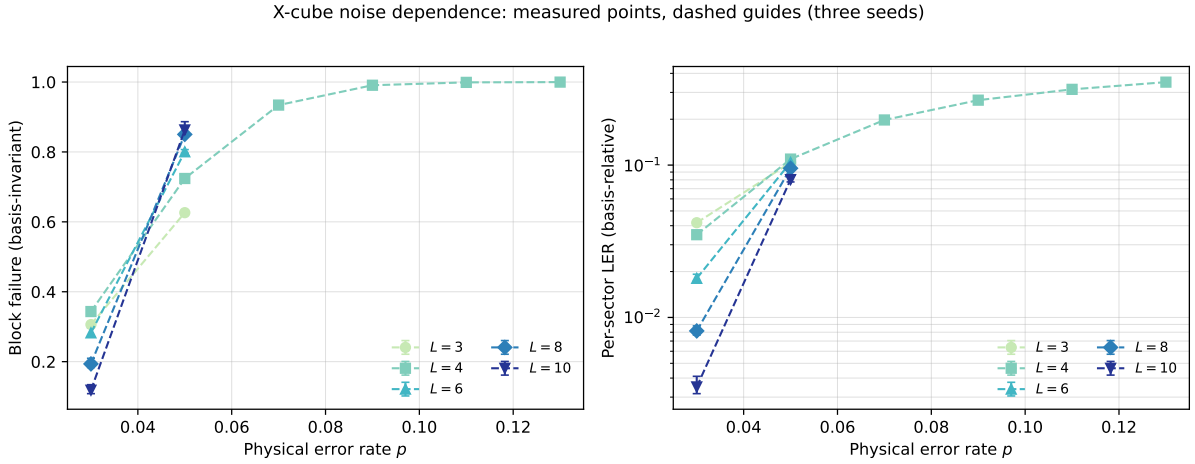


Figure 2: Measured learned-decoder error versus physical error rate (3-seed means and seed-bootstrap 95% intervals). Left: block failure. Right: mean per-sector error in the fixed logical basis. Points are measurements; dashed connectors are guides only. Except for $L = 4$, each curve has just two points, $p = 0.03$ and $p = 0.05$. No threshold is estimated from their interpolation.

Table 2: Head ablation at $L = 3$, $p = 0.03$ (seed 42). Global pooling ties trivial; structure-preserving heads win.

Head	Block failure	Per-sector LER
Trivial (floor)	0.7342	0.0846
Global average pool	0.7342	0.0847
Coordinate conv	0.3022	0.0420
AxisLineonHead	0.3026	0.0421

information lies outside its window. The geometric mechanism remains a hypothesis supported by the architecture and these restricted comparisons.

Predeclared win condition. The fixed-in-advance criterion (learned block-failure CI upper bound 0.6282 below trivial CI lower bound 0.8859 at $L = 3$, $p = 0.05$) is satisfied.

6 Related work

Neural decoders have been developed extensively for the surface and toric codes, including equivariant architectures (The END [6]; the 3D-toric equivariant ML decoder [7], which shows weight-sharing is necessary but not sufficient) and scalable CNNs for 3D/4D toric codes [8]. For fracton codes specifically, prior decoding work is *algorithmic*: Brown–Williamson give a parallelized matching+BP fracton decoder [5] (threshold $\approx 4.3\%$), and the optimal X-cube fracton-sector threshold under X-noise is $\approx 7.5\%$ [4]. Neural-network quantum states have been applied to fracton *ground states* [9] but not to decoding. Our contribution is an X-cube-specific neural decoder and its released finite-size benchmarks; we do not rely on a claim of priority. We benchmark it against the BW-naive implementation of Brown–Williamson's code-specific matching framework [5] described in §4, alongside the trivial floor and BP+OSD-0. We distinguish it from the parallel line of work that decodes high-rate qLDPC codes by *algebraic* means: Wills et al. [11] concatenate a quantum Reed–Solomon outer code over a bivariate-bicycle qLDPC inner code, decoded by list decoding and (Relay-)BP, with no learned component — a different code family *and* a non-neural method. Learned message-passing decoders for qLDPC and bivariate-bicycle codes also exist: Astra [10] uses a graph-neural/message-passing architecture on Tanner

graphs and reports extrapolation across code distances; it is complementary to our X-cube-specific line-aggregation head rather than a prior fracton decoder. Comprehensive surveys of the decoder landscape [12] confirm that machine-learning decoders and fracton-adjacent topics are each individually mainstream; our study focuses on their intersection in a particular code and noise model. The self-correcting-memory motivation follows the fracton literature (Haah's code [2] and the partial-self-correction line).

7 Discussion and limitations

The retained measurements demonstrate a learned decoder with improving finite-size block recovery from $L = 4$ through $L = 10$ at $p = 0.03$, together with monotone improvement in the fixed-basis per-sector metric from $L = 3$. They also show the limit of that observation: block failure worsens with size at $p = 0.05$. All experiments use perfect syndrome measurements and independent X noise; noisy repeated measurements and other noise channels are outside the present scope.

Objective and resource budget. Binary cross-entropy and validation selection optimize per-sector prediction, not the joint logical-class decision. Joint or structured logical prediction is a useful follow-up for block recovery. Weight sharing keeps the network parameter count independent of L, but models are trained separately at each (L, p), and processing a larger lattice requires more computation. We do not measure training-time or sample complexity scaling and do not demonstrate zero-shot transfer.

Baselines and statistics. BP+OSD-0 is one general-purpose configuration; higher-order OSD and other tuned variants are not benchmarked. BW-naive is a simplified implementation whose logical performance has not been validated against the published algorithm. Its poor large-size behavior must be audited before using it to support claims about code-specific matching. Large paired test-set differences establish gaps between the implemented decoders on the sampled distribution, not between all methods in those families. Three training seeds provide limited information about training variability; the reported seed-bootstrap intervals should be read alongside individual-seed results and test-shot counts rather than as precise population coverage estimates.

Next experiments. A dense common p grid across sizes would test the apparent reversal in block-failure ordering. A validated code-specific baseline and larger-size, matched-capacity head ablations would strengthen the performance and mechanism claims. The $L = 4$ sweep already shows the per-sector advantage over trivial narrowing from 0.110 versus 0.172 at $p = 0.05$ to 0.267 versus 0.274 at $p = 0.09$, and disappearing around $p = 0.11$–0.13; these are single-size comparisons, not threshold estimates. Transfer to 3D toric or other quantum codes would test the broader architectural idea, but the appropriate readout and baseline depend on each code's logical operators and syndrome geometry.

The classical motivation remains useful: a prior and a compact constraint description can support inference without specifying a unique microscopic answer. Here the acceptable ambiguity is precisely stabilizer equivalence. Recovering the correct logical class is the benchmark; constructing a passive or self-correcting memory would require additional physical results.

Reproducibility

All numbers are emitted by reproducible scripts. The main grid, ablation, and signal-existence oracle come from `reproduce_xcube_results.py` (output `data/xcube_L346_gen.json`, with

the $L \geq 8$ scaling rows in `data/xcube_L810_gen.json`); the high-p sweep from the same script (`data/xcube_L4_psweep.json`); and the BW-naive baseline from the consolidated $L \leq 6$ run plus the parallel $L \geq 8$ CPU run (`reproduce_xcube_bw_hpc.py`), with the standalone `data/xcube_bw_baseline.json` retained for provenance. Full provenance, including the per-cell statistics of App. A, is in `REPRODUCIBILITY.md`. In the family release, these artifacts are grouped with the X-cube and gauge-theorem code under the fracton module of the companion repository.

A Per-cell confidence intervals and McNemar counts

Table 3 reports the seed-bootstrap 95% confidence intervals on block failure for the learned and BW-naive decoders, and the paired-McNemar discordant counts (b, c) on identical test shots (b = shots the learned decoder gets right while the comparison decoder gets them wrong; c = the reverse). All paired McNemar tests are overwhelmingly significant; the continuity-corrected χ^2 statistic (or an exact binomial test) gives $p < 10^{-12}$ in every tabulated paired comparison (learned-vs-BW-naive is tabulated for $L \leq 6$ only; at $L \geq 8$ BW-naive is compared by aggregate failure rate, not paired shot-by-shot). BP+OSD-0 is evaluated on the smaller cost-limited subsets noted in Table 1 (2,000 shots/seed through $L = 8$, 300 at $L = 10$); even at the smallest subset its block-failure Wilson 95% interval stays far from the learned value (e.g. at $L = 10, p = 0.03$, BP+OSD-0 0.978, Wilson $[0.966, 0.986]$ on 900 shots, vs. learned 0.118), so the reported gaps vastly exceed the baseline sampling error.

Table 3: Per-cell block-failure 95% CIs and paired-McNemar discordant counts (b, c) for the learned decoder against trivial, BP+OSD-0, and BW-naive (the last for $L \leq 6$; — at $L \geq 8$, where BW-naive fails near-totally and is compared by aggregate failure rate). $L \geq 8$ from task-parallel per-cell runs.

		Learned bf	BW-naive bf	McNemar (b, c)		
L	p	[95% CI]	[95% CI]	vs. trivial	vs. BP+OSD-0	vs. BW-naive
3	0.03	0.306 [0.303, 0.308]	0.391 [0.387, 0.394]	(53294, 1794)	(511, 99)	(12648, 2499)
3	0.05	0.626 [0.624, 0.628]	0.703 [0.699, 0.705]	(33405, 2079)	(499, 115)	(12663, 3470)
4	0.03	0.343 [0.343, 0.344]	0.682 [0.681, 0.685]	(69620, 1154)	(988, 257)	(43529, 2852)
4	0.05	0.724 [0.720, 0.728]	0.937 [0.936, 0.938]	(31555, 719)	(859, 146)	(26963, 1399)
6	0.03	0.282 [0.272, 0.296]	0.935 [0.934, 0.936]	(85691, 31)	(2067, 218)	(78726, 369)
6	0.05	0.800 [0.796, 0.807]	0.999 [0.999, 0.999]	(23950, 11)	(1108, 22)	(23854, 30)
8	0.03	0.194 [0.183, 0.210]	0.993 [0.992, 0.993]	(72574, 0)	(4067, 23)	—
8	0.05	0.850 [0.842, 0.856]	1.000 [1.000, 1.000]	(13485, 0)	(905, 0)	—
10	0.03	0.118 [0.108, 0.134]	1.000 [0.999, 1.000]	(79351, 0)	(792, 1)	—
10	0.05	0.863 [0.851, 0.887]	1.000 [1.000, 1.000]	(12337, 0)	(117, 0)	—

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